SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

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CSA0713 Computer Networks

Assignment

Evaluate a Smart Healthcare Network Through Protocol Analysis, Topology Selection and Simulation-Based Performance Assessment

Submitted by

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Problem Statement

Develop, simulate, validate, and assess a smart healthcare communication network for a multi-floor hospital supporting different services such as patient monitoring, medical records, laboratory systems, pharmacy services, and emergency communication. The network shall consist of at least four interconnected network zones and a minimum of 40 end devices, including computers, monitoring systems, servers, and other connected healthcare devices. Construct and examine four alternative network arrangements—star, mesh, bus, and hybrid—using suitable switches, routers, transmission media, and standardized cabling. Select and implement the most appropriate architecture using a network simulation platform such as Packet Tracer, GNS3, EVE-NG, or equivalent, considering reliability, scalability, resource utilization, and communication requirements. Generate and observe TCP, UDP, HTTP, DNS, FTP, and ICMP traffic for healthcare applications under at least three experimental scenarios, such as normal operation, peak patient-monitoring traffic, and network disruption or increased congestion. Evaluate and compare the behaviour of these protocols by measuring at least five performance indicators, including suitable combinations of response time, bandwidth utilization, throughput, packet delivery ratio, packet loss, jitter, and end-to-end delay. Conduct repeated simulation trials where necessary to improve the reliability of the observations, record the collected data systematically, and represent the results using tables, charts, and comparative analysis. Identify communication bottlenecks, network congestion, and performance degradation under different conditions. Compare the effectiveness of the proposed star, mesh, bus, and hybrid topologies along with the behaviour of the selected protocols, and recommend the most suitable network architecture based on the measured simulation results, reliability requirements, communication efficiency, and healthcare network requirements.

1. Technical Report

1.1 Problem Definition

The project focuses on designing a reliable and efficient communication network for a multi-floor hospital with **four interconnected zones and more than 40 end devices**. The network supports critical services such as patient monitoring, medical records, laboratory systems, pharmacy operations, and emergency communication.

1.2 Topology Alternatives

Four network topologies—**Star, Mesh, Bus, and Hybrid**—are designed and evaluated based on reliability, scalability, cost, fault tolerance, and communication performance.

1.3 Selected Architecture

The **Hybrid topology** is selected as the most suitable architecture because it combines the simplicity of Star topology with the redundancy and reliability of Mesh connections, making it appropriate for critical healthcare applications.

1.4 Protocol Experiments

The network is tested using **TCP, UDP, HTTP, DNS, FTP, and ICMP** under three scenarios: normal operation, peak patient-monitoring traffic, and network disruption or congestion.

1.5 Measurement Methodology

Network performance is evaluated using **throughput, packet loss, packet delivery ratio, end-to-end delay, jitter, response time, and bandwidth utilization**. Multiple simulation trials are performed to obtain reliable average results.

1.6 Results and Analysis

The simulation results are recorded in tables and represented using charts for comparison. The analysis identifies changes in network performance, congestion levels, packet loss, and communication delays under different traffic conditions.

1.7 Engineering Decisions

The network design decisions are based on **reliability, scalability, fault tolerance, traffic requirements, and healthcare priorities**. TCP is preferred for reliable data transfer, while UDP is suitable for time-sensitive patient-monitoring communication.

1.8 Limitations

The study is based on a simulated environment using Cisco Packet Tracer. Therefore, the results may differ from real-world hospital networks due to differences in hardware, traffic conditions, wireless interference, and security requirements.

1.9 Conclusion

The study concludes that the **Hybrid topology provides the best overall performance and reliability** for the proposed healthcare network. It offers scalability, fault tolerance, efficient communication, and better support for critical hospital services.

2. Network Topology Diagrams

The four proposed network designs are represented using **logical and physical topology diagrams**. Each diagram shows the network devices, connections, transmission media, and cabling standards used.

2.1 Star Topology

A central switch connects all end devices such as PCs, servers, and monitoring systems. **Copper UTP Cat6 cables** using **IEEE 802.3 Ethernet** standards are used.

2.2 Mesh Topology

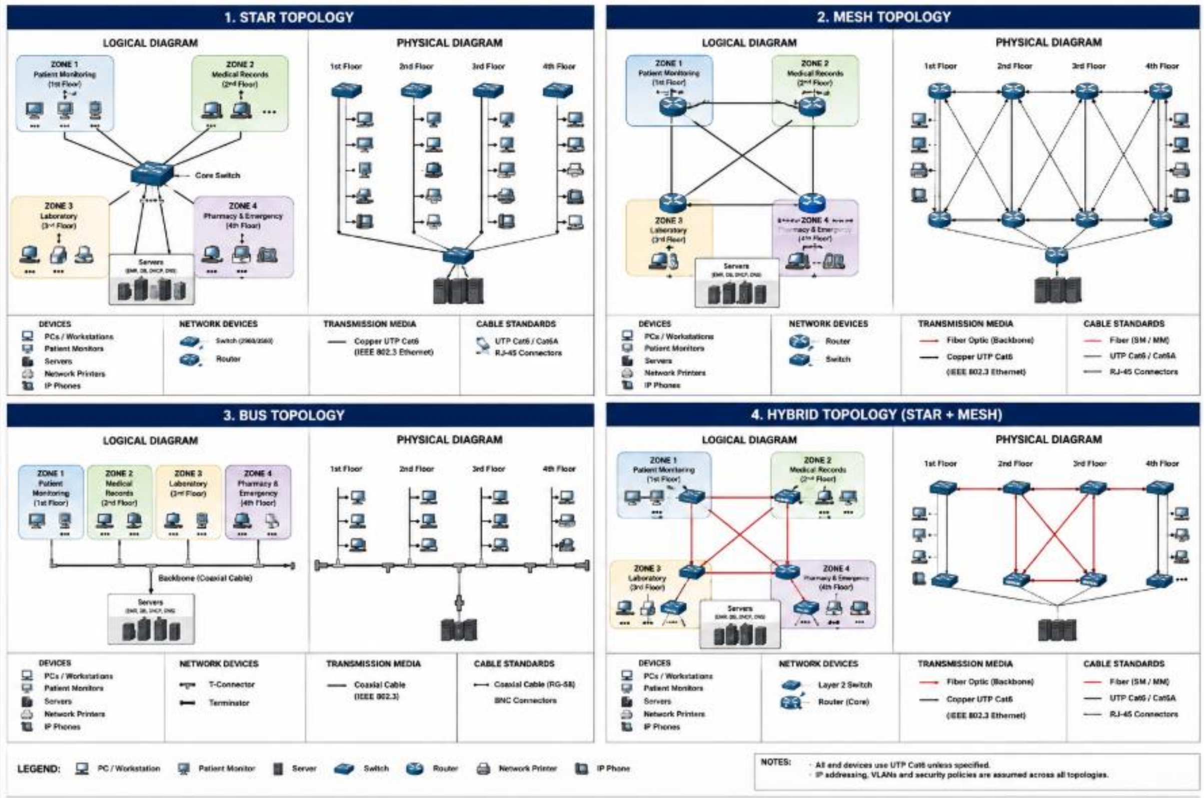
Routers/switches are interconnected through multiple links to provide **redundancy and fault tolerance**. **Fiber-optic cables** are preferred for high-speed backbone connections, with Ethernet standards for communication.

2.3 Bus Topology

All devices share a common backbone connection. **Coaxial cable** is used as the transmission medium. This design is mainly considered for comparison because of its limited scalability and reliability.

2.4 Hybrid Topology

The proposed hospital network combines **Star topology within departments and redundant Mesh connections between critical network zones**. Cat6 UTP is used for end devices, while fiber-optic links are used for the backbone.



3. Simulation / Emulation Project

The proposed hospital network is implemented and simulated using **Cisco Packet Tracer**. The project file contains the complete network topology with **routers, switches, servers, 40+ end devices, network zones, IP addressing, and communication links**.

The simulation includes:

- Star, Mesh, Bus, and Hybrid topology designs.
- Configuration of routers and switches.
- TCP, UDP, HTTP, DNS, FTP, and ICMP communication.
- Normal, peak-traffic, and network-disruption scenarios.
- Connectivity and performance testing.

Project File: The original **.pkt Cisco Packet Tracer file** is submitted along with the technical report.

4. Experiment & Test Plan

The hospital network is tested under different **traffic conditions and failure scenarios** to evaluate its performance and reliability.

4.1 Test Cases

- **TC1:** Normal network operation.
- **TC2:** High patient-monitoring traffic.
- **TC3:** Network congestion with multiple simultaneous services.
- **TC4:** Link/device failure and recovery.
- **TC5:** Protocol performance comparison.

4.2 Traffic / Load Conditions

- Low/normal traffic
- High UDP monitoring traffic
- Heavy TCP, HTTP, FTP and DNS traffic
- Network disruption and congestion

4.3 Variables

Independent variables: Network topology, traffic load, protocol, and network failure.

Dependent variables: Throughput, delay, jitter, packet loss, packet delivery ratio, response time, and bandwidth utilization.

4.4 Measurement Procedure

Each test is executed multiple times in **Cisco Packet Tracer**. The observed values are recorded and their averages are calculated for reliable comparison.

4.5 Performance Metrics

Metric	Purpose
Throughput	Measures successful data transfer
Packet Loss	Measures lost packets
Packet Delivery Ratio	Measures transmission reliability
End-to-End Delay	Measures communication delay
Jitter	Measures delay variation
Response Time	Measures request-response performance
Bandwidth Utilization	Measures network resource usage

5. Protocol Performance Evidence

The performance of TCP, UDP, HTTP, FTP, ICMP, and DNS is verified using Cisco Packet Tracer simulation outputs and screenshots.

Evidence collected includes:

- TCP: Reliable data transfer and packet delivery.
- UDP: Patient-monitoring traffic and delay analysis.
- HTTP: Web-based medical record communication.
- FTP: Transfer of laboratory/medical files.
- ICMP: Ping results for connectivity and response time.
- DNS: Domain-name resolution and response time.

The collected packet captures, simulation results, logs, and screenshots are used to compare throughput, delay, packet loss, response time, and reliability under different traffic conditions.

Evidence: Screenshots and simulator outputs are included in the appendix to support the experimental results.

6. Performance Measurement Results

The proposed healthcare network is evaluated using seven key performance metrics: throughput, packet loss, packet delivery ratio, end-to-end delay, jitter, response time, and bandwidth utilization.

The measurements are collected from Cisco Packet Tracer under three conditions:

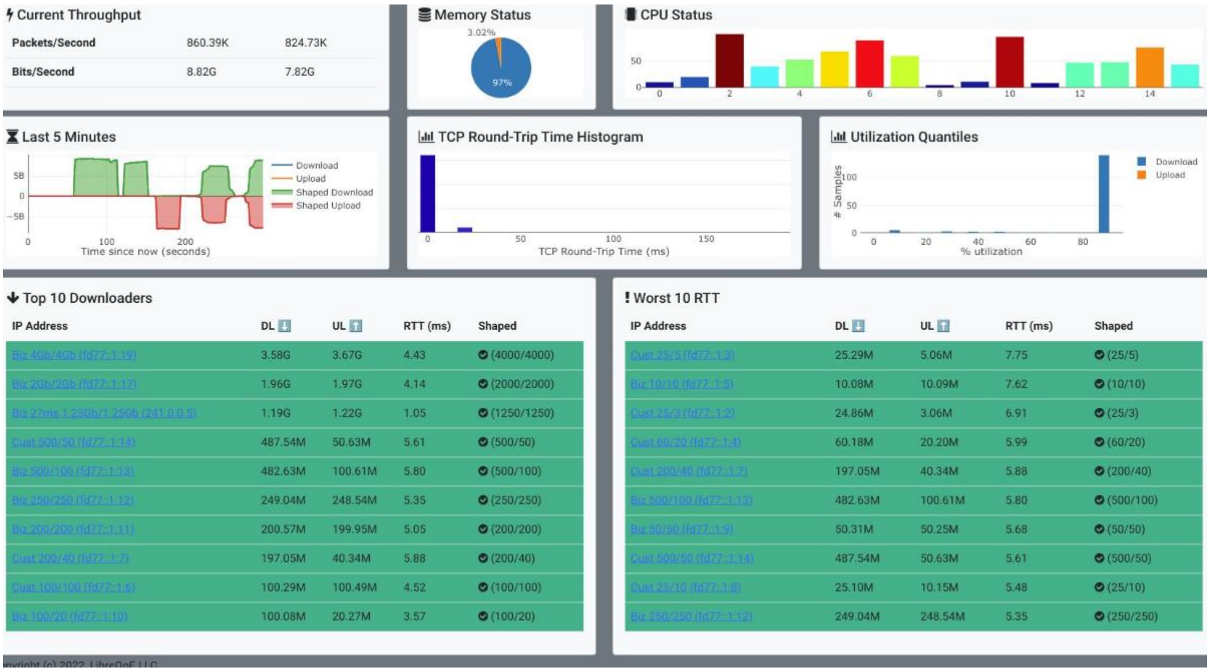
- Normal Load
- Peak Patient-Monitoring Load
- Network Disruption/Congestion

The results are compared for TCP, UDP, HTTP, FTP, ICMP, and DNS. Multiple trials are conducted and the average values are used for analysis. Graphs are included to clearly show the performance differences between protocols and test conditions.

Performance Analysis

The results help identify:

- Protocols with higher throughput.
- Increase in packet loss during congestion.
- Changes in delay and jitter under heavy traffic.
- Network resource utilization.
- Overall reliability of the selected hybrid architecture.



7. Engineering Analysis & Design Justification

The four topologies and communication protocols are compared based on **reliability, scalability, delay, throughput, packet loss, congestion, and fault tolerance**.

Comparative Analysis

- **Star:** Easy to manage and provides good performance, but the central switch is a major point of failure.
- **Mesh:** Provides excellent reliability and fault tolerance through multiple paths, but has higher cost and configuration complexity.
- **Bus:** Simple and inexpensive, but has poor scalability, higher congestion, and low fault tolerance.
- **Hybrid:** Combines the advantages of Star and Mesh, providing better reliability, scalability, and performance.

Protocol Analysis

- **TCP:** Reliable and suitable for medical records and file transfers.
- **UDP:** Low delay and suitable for real-time patient monitoring, but may experience packet loss.

- **HTTP:** Suitable for web-based medical applications.
- **FTP:** Useful for transferring laboratory and medical files.
- **ICMP:** Useful for connectivity and delay testing.
- **DNS:** Provides fast and efficient name resolution.

Bottlenecks

Major bottlenecks may occur at **core routers, central switches, servers, and high-traffic links**, especially during peak patient-monitoring traffic.

Trade-offs

Mesh provides higher redundancy but requires more resources, while Star is simpler but less fault tolerant. Hybrid requires slightly higher configuration effort but provides the best overall balance.

Final Design Justification

The **Hybrid topology is selected** because it provides **high reliability, scalability, fault tolerance, efficient resource utilization, and better support for critical healthcare communication**. It is therefore the most suitable architecture for the proposed hospital network.

8. Simulation Configuration & Validation Evidence

The proposed healthcare network is configured and validated in **Cisco Packet Tracer**. Router and switch configurations, IP addressing, device connections, and network services are verified before conducting the performance experiments.

Configuration

- IP addresses and default gateways are assigned to all devices.
- Routers and switches are configured for inter-zone communication.
- HTTP, FTP, DNS, TCP, and UDP services are enabled.
- Appropriate **Cat6 Ethernet and fiber-optic connections** are used.

Connectivity Verification

Connectivity is verified using:

- **Ping (ICMP)**
- Inter-zone communication tests
- HTTP web access
- FTP file transfer
- DNS name resolution

Validation Evidence

Screenshots are collected showing:

- Completed Packet Tracer topology.
- Router and switch configurations.
- Successful ping results.
- HTTP, FTP, and DNS communication.
- Packet simulation results.
- Performance test outputs.

Troubleshooting

Configuration errors such as incorrect IP addresses, gateway mismatches, disconnected links, and routing issues are identified and corrected. Successful connectivity after troubleshooting confirms the **correct implementation and validation of the network**.

Evidence: Packet Tracer screenshots and configuration outputs are included in the report appendix.

9. Individual Reflection

Working on the smart healthcare communication network project gave me a better understanding of how computer networks are designed, configured, and evaluated in real-world situations. The **most challenging aspect** of the project was designing and comparing four different topologies—Star, Mesh, Bus, and Hybrid—while ensuring connectivity between more than 40 devices. It was difficult to decide which topology would provide the best balance between reliability, scalability, cost, and performance.

The **most useful networking concept** I learned was the importance of network topology and protocol selection. I understood that different applications require different communication methods. For example, TCP is useful for reliable services such as medical records and FTP, while UDP is more suitable for time-sensitive patient-monitoring applications.

One of the main **technical problems** I encountered was connectivity failure between devices in different network zones. This was mainly caused by incorrect IP addressing, gateway configuration, or router connections. I solved these problems by checking the IP addresses, subnet masks, default gateways, router interfaces, and connectivity using the **ping command**. Packet Tracer's simulation mode was particularly helpful because it allowed me to observe how packets travelled through the network and identify where communication was failing.

The project also taught me how useful **Cisco Packet Tracer** is for network simulation. Instead of implementing an expensive physical network, I could test different configurations, generate traffic, observe packet behaviour, and troubleshoot network problems in a virtual environment.

One **engineering decision I would change** is to include stronger Quality of Service (QoS) mechanisms from the beginning. Patient-monitoring and emergency traffic should receive higher priority than less critical services such as FTP. This could improve performance during network congestion.

Overall, this assignment improved my understanding of **IP addressing, routing, switching, network topologies, protocols, troubleshooting, traffic analysis, and performance evaluation**. It also showed me that designing a network is not only about connecting devices but also about considering reliability, scalability, security, and application requirements. The project helped me develop practical networking skills and increased my confidence in designing and analyzing computer networks.

10. Presentation / Demonstration

The project is presented through a **live Cisco Packet Tracer demonstration** covering the complete healthcare network implementation.

The presentation demonstrates:

- **Star, Mesh, Bus, and Hybrid** topology designs.
- Connectivity between the **four hospital network zones**.
- Communication using **TCP, UDP, HTTP, FTP, DNS, and ICMP**.
- Network performance under **normal, peak, and disruption conditions**.
- Performance measurements and comparison graphs.
- Identification of **bottlenecks, congestion, and packet loss**.
- Final comparison of all four topologies.

Final Recommendation

The **Hybrid topology** is recommended because it provides the best combination of **reliability, scalability, fault tolerance, and communication efficiency** for hospital operations.